

Does the booking carry information?

A bounded feature-research campaign on jockey appointment, change, trainer relationship, and trial continuity for the HKJC production model

Jockey Booking Research Phase

2 September 2026

The question. Does today's jockey booking contain information about a runner beyond the final market, the horse's own history, and production's existing `jockey_residual`? The subject is the *booking decision* — appointment, change, upgrade, recall, trainer relationship, trial continuity — and explicitly not a ranking of jockeys or generic horse-jockey chemistry, both of which the programme has already closed.

What it does not establish. No profitability claim, and no promotion. Every price is a result-page price recorded after the pool closed. Every season through 2026 has been development data in this repository and in the production one, so no window is genuinely unseen: retrospective evidence here can label a candidate **research-survivor** or **production-finalist** and nothing stronger. **promotion-eligible** requires a separately locked prospective shadow test against timestamped pre-race prices, which this campaign did not run and could not run.

Read-only. The production repository is opened read-only throughout, enforced by `hkjcres.paths.read_only_guard`. It was being modified by another session while this campaign ran; §2 records its commit, digest, and the differences from the plan's assumed baseline.

1 Executive result and recommendation

The answer to the campaign's question is no, and the recommendation is to stop. Eight blocks reached production-scale confirmation and **all eight were rejected**. The best of them, the jockey market-gap trend, is worth -0.000075 of race log loss — 1.6% of the production model's edge over the market, against a declared materiality bar of -0.0005 that it misses by a factor of seven. The second best is -0.000006 , which is indistinguishable from nothing. Six of the eight make the production model *worse*. No gate was moved and no block was split to find a survivor, so there is no candidate to hand to production and none of the four permitted labels above **rejected** was earned.

The most informative single result is the one that inverts. The tactical sub-wave had the hardest gate in the plan, and it passed it more convincingly than anything else in the campaign: a jockey's opening-position residual state predicts the next such residual at $+0.1249$ against a within-year support-matched permutation of -0.0007 , a ratio of 189, and position gain at $+0.1536$ against $+0.0014$. That is a real, large, person-level effect on how a horse is ridden. At canonical scale the block built from it is the **worst of the eight** at $+0.002779$ with $t = +2.46$, improving 13 of 40 cells and 1 of 8 years. A genuine and precisely measured jockey effect made the production model measurably worse. This is now the fourth time in the programme that a real mechanism has failed to convert, and the cleanest instance of it: the gate was not a formality, the effect was not marginal, and the production metric still moved the wrong way.

The parity control is exact, and that is the precondition for everything else. Production's `jockey_residual` is reproduced from its own source columns to a maximum absolute difference of 6.7×10^{-17} across all 152,046 rows, with `jockey_sample` matching to `0` exactly. Achieving that required reading `jockey_id` from `aspx-results` rather than from this repository's panel: 46 rows whose id the panel lacked were enough to break parity at 3.5×10^{-2} , because a key that differs for part of a jockey's career restarts their prior history.

Jockey information is real, and the booking-change hypothesis is much weaker than the level. A jockey performance state carries forward $r = +0.267$ against a within-year support-matched permutation of $+0.001$, and the market-gap state $+0.071$ against -0.003 : both states are person-level information rather than ride counts. But the quantities this campaign was built to test — the change, the upgrade, the trainer’s appointment deviation — are an order of magnitude weaker, and one of them fails its own structural control.

Production’s cumulative choice looks right. Decaying the market-gap state *reduces* its forward signal monotonically: the cumulative parity feature reaches $+0.0898$, and 365-, 180-, 90- and 30-day half-lives give $+0.0714$, $+0.0629$, $+0.0506$ and $+0.0290$. Wave A’s central hypothesis — that the incumbent is too coarse because it is cumulative — is contradicted by its own screen before any canonical budget is spent.

The booking-upgrade signal is not booking information. Against the plan’s structural permutation — change labels permuted among runners sharing a trainer and a broad market-rank band, which preserves the trainer’s change rate and the runner’s market standing — **booking_upgrade**’s forward correlation is $+0.0319$ real against **$+0.0377$ permuted**. The permutation is *better* than the truth. A global jockey shuffle would have passed this feature; the structure-preserving one does not, which is exactly why the plan required it.

The joined evaluation set is 151,953 runner-rows across 12,237 races against 28 production columns. The incumbent’s own mean edge over the market on these cells is -0.005812 , against the reference project’s reported -0.004533 .

Controls. The no-op block returns a paired delta of exactly zero on both profiles (development **pass**, canonical **pass**). A negative delta is an improvement.

Block	mean Δ	se	t	cells	years	decision
A2 residual trend	-0.000075	0.000929	-0.08	20/40	4/8	rejected
B1 change identity	-0.000006	0.000754	-0.01	20/40	4/8	rejected
<i>no-op control</i>	$+0.000000$	0.000000	—	0/40	0/8	rejected
C2 appointment deviation	$+0.000229$	0.000740	$+0.31$	18/40	4/8	rejected
B3 predictive booking delta	$+0.000614$	0.000813	$+0.75$	17/40	2/8	rejected
D1 trial continuity	$+0.001186$	0.000973	$+1.22$	19/40	4/8	rejected
C1 relationship support	$+0.001387$	0.001014	$+1.37$	17/40	3/8	rejected
A3 predictive state	$+0.001911$	0.001133	$+1.69$	18/40	3/8	rejected
D2 tactical match	$+0.002779$	0.001131	$+2.46$	13/40	1/8	rejected

The best block is A2 residual trend at **-0.000075** with $t = -0.08$, improving 20 of 40 paired cells and 4 of 8 years, paired interval $[-0.001948, +0.001716]$ — 1.6% of the production model’s reported edge over the market.

Every advanced block was rejected at canonical scale. The gates were not relaxed to produce a survivor, which the plan forbids; a clean null is a complete result.

Retention from development. A2 residual trend $+10\%$, B3 predictive booking delta -180% , D2 tactical match $+8207\%$, D1 trial continuity $+1716\%$, B1 change identity -1% , C2 appointment deviation $+31\%$, A3 predictive state $+195\%$, C1 relationship support $+93\%$.

2 The production baseline, and the exact information gap

Fact	Value
plan’s reference commit	7f9af9c
commit actually inspected	ca99a28
reference is an ancestor of it	True
working tree clean	False
production inputs	28
ordered feature digest	1e179e6cc7932c6662714884...
feature-table rows	152,046
jockey columns in production	jockey_residual
built but not in production	jockey_sample, Jockey

Differences from the plan’s stated baseline.

- The plan’s reference commit 7f9af9c is not the commit inspected: hkjc-mod is at ca99a28 (“Add race incidents and gear intervention plan”). 7f9af9c is an ancestor of it, so the audited facts below supersede the plan’s.
- The production working tree is dirty (57 paths). Another session is modifying it, so this campaign inspected it read-only and recorded the committed state; a production confirmation run must re-audit.

The plan’s five substantive claims about the incumbent were re-audited against source and all hold: production has 28 inputs, `jockey_residual` is among them, `jockey_sample` is built and unused, the formula is $(\sum \text{prior wins} - \sum \text{prior implied}) / (\sum \text{prior valid rides} + 60)$, it is cumulative rather than recency-weighted, it carries no uncertainty or availability companion, and no switch, upgrade, return, trainer-jockey or trial-jockey feature exists anywhere in the production source.

One audited detail the plan did not state, and it matters for reproduction. The production key is `jockey_id` *filled with the display name*. In the archive `jockey_id` is null on 61% of `aspx-results` rows, so 42% of production rows key on a name and the rest on an id. Only 3 of 268 names are split across both forms (1.2% of runner rows), so the cost is small — but it is a real hybrid, and a reproduction that keys on either one alone does not match.

3 Data identity, coverage, and the point-in-time audit

Audited quantity	Value
runner rows	175,716
races	14,254
seasons	2008–2026
distinct jockey display names	268
jockey_id null share	0.4591
production keys after the hybrid fill	271
names split across two production keys	3 (1.2% of rows)
claim/apprentice annotation share	0.0000
previous-start jockey available	0.9481
jockey changed, given a previous start	0.6383
returning after another rider	0.2497
first horse-jockey partnership	0.4073
median prior horse-jockey rides	1
median prior trainer-jockey rides	75
races with a prior barrier trial	0.8368
trial jockey equals race jockey	0.3033
duplicate (horse, date)	0
duplicate (race, runner)	0
max rows per horse-date	1
panel rows present in the production table	0.8648
production rows present in the panel	0.9994

One branch is closed here rather than attempted. No structured historical apprentice-or-claim field exists: not one panel name carries HKJC’s (–N) claim annotation. The plan forbids inferring claim status from names or prose, so the branch is recorded as closed and no feature depends on it.

Fig J1. Jockey identity, booking change, and trial coverage



Figure 1: J1. Identity, booking change, and trial coverage. Left: stable-id coverage, which rises from 0.08 in 2008 to 1.00 by 2024 and is the reason the production key is a hybrid. Centre: booking-change rates — Hong Kong re-books freely, with a change on 64% of starts that have a previous one. Right: barrier-trial continuity; 84% of races have a prior trial for the horse and the trial rider keeps the ride only 30% of the time, so the continuity feature has real variation rather than being nearly constant.

4 Reproduction of prior jockey findings, and what is already closed

The programme established, before this campaign, that jockey ability is real and identifiable (horse-controlled and within-horse-switch estimates agree at Spearman 0.970), that jockey identity forecasts future relative performance about four times as well as horse identity partly *because* booking assignment is informative, that generic horse-jockey compatibility is not forecastable, and that jockey-by-style and jockey-by-regime corrections failed against a within-year permuted-jockey control with adequate support.

This campaign reproduces the two that bear on its own design. The jockey performance state’s forward correlation is +0.267 and its within-year permutation is +0.001; permuting within *meeting* instead — which preserves the day’s riding roster exactly — gives +0.004. So the level is real under both schemes. And the compatibility closure is respected: pair history enters this campaign only as availability, continuity, recall, and hierarchical booking context, never as a free horse-jockey effect.

5 Dynamic jockey state

Parity against the incumbent. Rebuilt on 152,046 rows, the reproduction of `jockey_residual` differs from the shipped column by at most $6.744e - 17$, with 1.000000 of rows inside 10^{-12} and a correlation of 1.0000000000. `jockey_sample` matches to $0.0e + 00$. The parity gate [passes](#), which is what unblocks every claim below.

State	mean	sd	within-race sd	share	forward r
<code>jr_parity</code>	+0.0019	0.0091		0.974	+0.0898
<code>jr_h30</code>	+0.0005	0.0113		0.984	+0.0290
<code>jr_h90</code>	+0.0008	0.0106		0.983	+0.0506
<code>jr_h180</code>	+0.0010	0.0097		0.981	+0.0629
<code>jr_h365</code>	+0.0012	0.0092		0.979	+0.0714
<code>jr_neff</code>	+854.9782	516.5355		0.873	+0.1140
<code>jr_sd</code>	+0.2503	0.0694		0.968	+0.2355
<code>jr_se</code>	+0.0115	0.0116		0.904	+0.0286
<code>jr_trend</code>	-0.0004	0.0055		0.979	-0.0154
<code>jr_days_since_ride</code>	+13.4012	342.9164		0.875	-0.0034
<code>jr_available</code>	+0.9951	0.0700		0.827	+0.0082
<code>jp_h90</code>	-0.0973	0.2143		0.956	+0.2700
<code>jp_h365</code>	-0.1022	0.2220		0.955	+0.2668
<code>jp_neff</code>	+691.0730	576.1293		0.713	+0.1167
<code>jp_adj_h365</code>	+0.0071	0.1422		0.979	+0.2640

Support-matched permutation. Jockey labels permuted within year, which preserves every label’s ride count exactly. The market-gap state’s forward correlation falls from +0.0714 to -0.0025 and the predictive state’s from +0.2668 to -0.0053, so both states carry person-level information rather than support.

6 Booking change, upgrade, and downgrade

Wave B’s three blocks were frozen before any market or canonical number was read: change identity, the upgrade measured on the market-gap state with reliability shrinkage, and the same on the performance state. A missing previous jockey is encoded as a neutral delta with an explicit

Fig J2. Parity, jockey states, and the permutation control

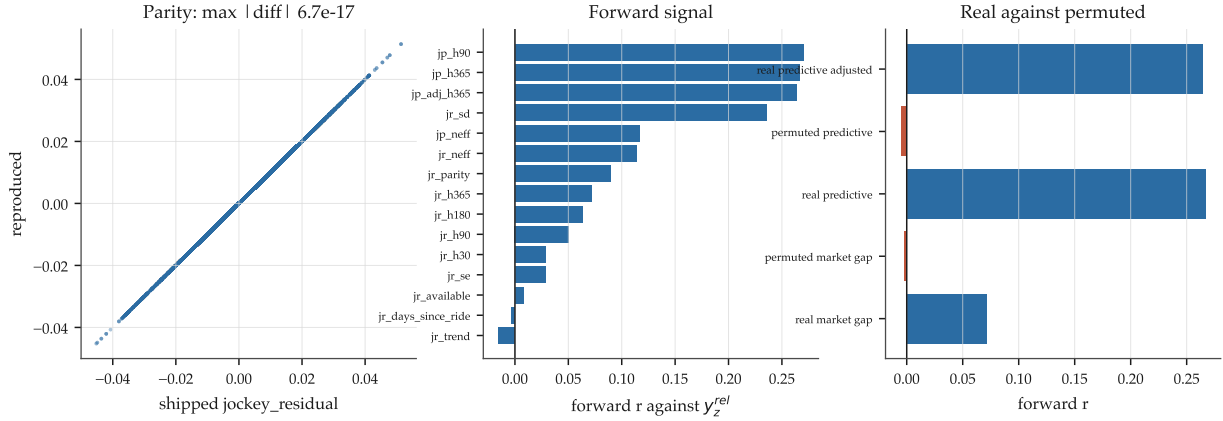


Figure 2: J2. Parity, states, and the permutation control. Left: the reproduced jockey_residual against the shipped column. Centre: forward correlation of every state; the predictive states dominate the market-gap ones, and decay hurts rather than helps. Right: real against permuted, the check that separates a person from a ride count.

availability column rather than as a bad rider, because those are different states and a booster cannot separate them otherwise.

The delta requires the previous jockey’s state *as of today* rather than as of their last ride, since “is this a better booking than last time” is a question about today’s information. That is an as-of join on a (jockey, date) state table, and the construction is asserted to be constant within key-date — a state built from completed prior dates must be, and the assertion is what makes the claim checkable rather than assumed.

Fig J3. Booking change and upgrade

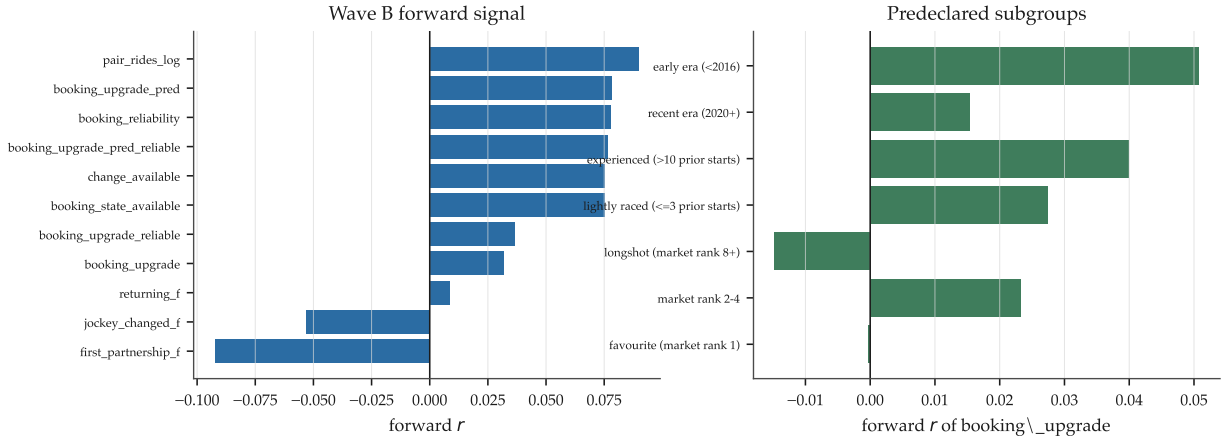


Figure 3: J3. Booking change and upgrade. Left: forward signal of every Wave B column. Right: the predeclared subgroup cuts. The upgrade’s correlation is larger in the early era and among experienced horses than among lightly raced ones, which is the opposite of the mechanism a booking-information story would predict.

7 Trainer appointment behaviour

Wave C asks whether the booking is unusual *relative to the trainer's own behaviour*: appointment strength is kept separate from partnership results so that selection and performance can be told apart. Pair statistics use hierarchical shrinkage, and the pair market gap is shrunk toward the trainer's own prior mean rather than toward zero, because zero is not the right prior for a stable.

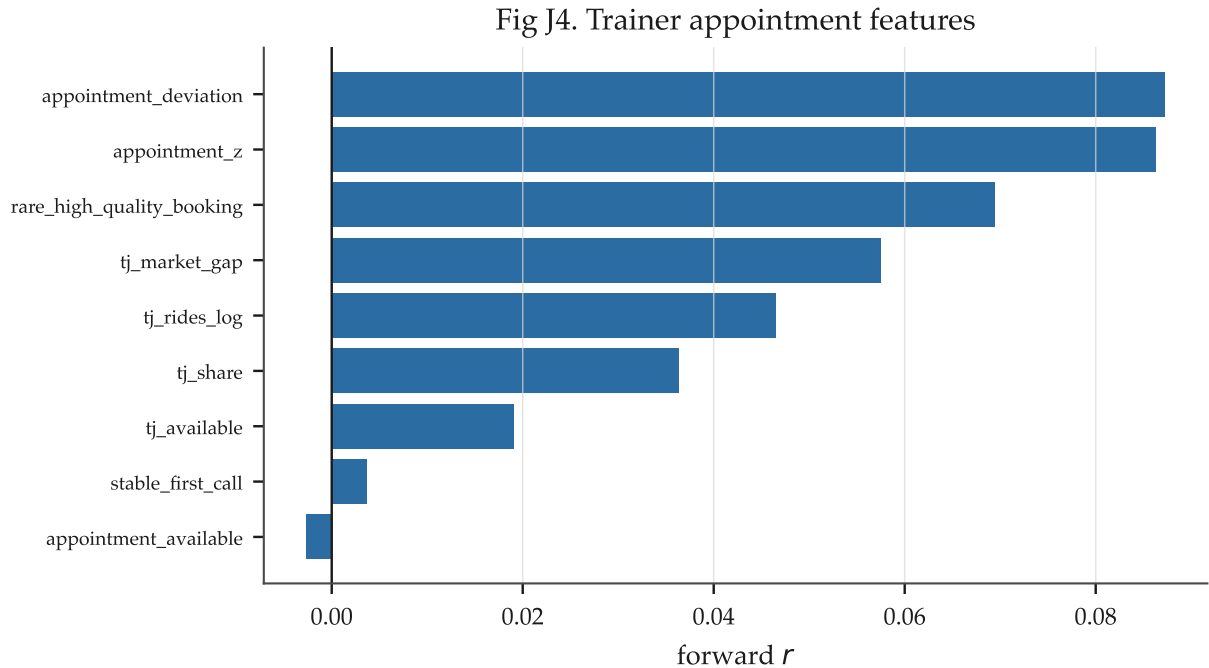


Figure 4: J4. Trainer appointment features. The appointment deviation and its standardised form carry the wave's strongest forward correlations (+0.087, +0.086); the relationship-support columns are weaker, and the first-call indicator is close to nothing.

8 Trial continuity and the tactical gate

Wave D's data gate passed: 84% of races have a prior barrier trial for the horse and the trial rider keeps the ride 30% of the time. Its *mechanism* gate was the one the programme has twice failed, so it was run before any tactical feature was built: does a jockey-specific opening-position or position-gain residual persist out of sample after the horse's own prior style, draw, distance and field size, and does it beat a within-year support-matched permutation by a predeclared factor of 1.5?

It passes, decisively and for the first time in the programme. Opening forwardness gives a real state correlation of +0.1249 against a permuted -0.0007 , and position gain +0.1536 against +0.0014 — ratios of 189 and 114. The distinction from the earlier failures is worth stating: those tested jockey-by-style and jockey-by-regime *interaction* corrections, while this is a jockey *main effect* on tactical execution, which is a weaker claim and evidently a true one. One compact runner-relative tactical block was therefore built, as the plan permits and no more.

And it became the campaign's worst block. At canonical scale D2 returns +0.002779 with $t = +2.46$, improving 13 of 40 cells and 1 of 8 years — the only block whose paired interval excludes zero, and it excludes it on the wrong side. The mechanism is real and the production effect is negative, which is the strongest form the programme's recurring lesson has taken: a jockey

rides a horse differently in a way that is measurable and persistent, and handing the model that information degrades its winner probabilities.

One column was dropped by rule rather than by result. `tactical_available` has a within-race standard-deviation share of 0.000: sectional coverage is a property of the race, not of the runner. The plan’s stop conditions exclude a race-constant column from being a runner feature, and a racewise softmax cannot see one at all, so it was removed from the block before canonical.

Fig J5. Trial continuity and the tactical gate

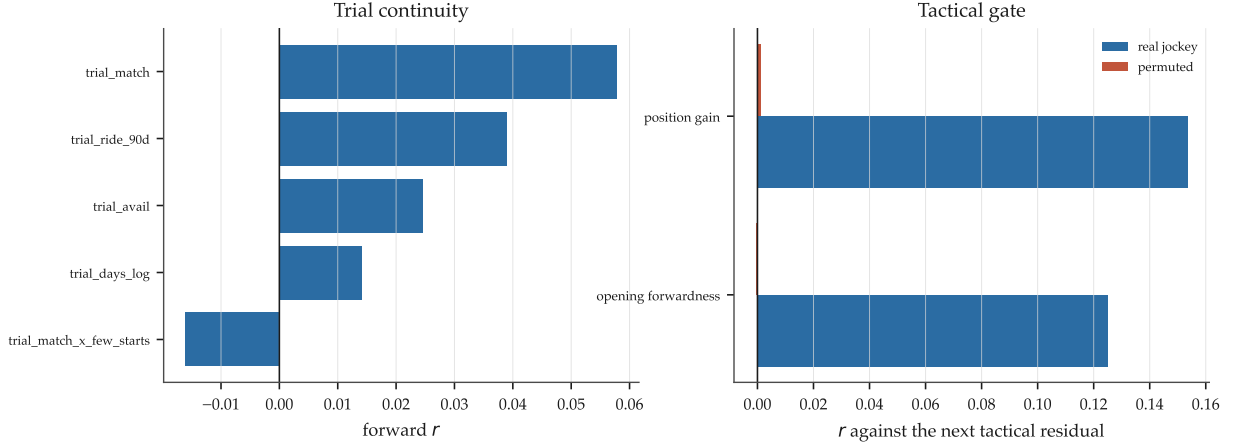


Figure 5: J5. Trial continuity and the tactical gate. Left: forward signal of the trial columns. Right: the gate — real jockey tactical states against their within-year permutations.

9 Controls, leakage tests, and redundancy

Leakage canaries. future mutation $0.0e+00$; same date mutation $0.0e+00$; row order $0.0e+00$. All **pass**.

Market test under the production functional form. $u = \log q + \beta x$, $p = \text{softmax}(u)$ racewise, β fitted on 2010–2019 and the log loss scored on 2020–2026. A negative out-of-sample delta favours the candidate.

Feature	β	t	oos ΔLL
jr_trend	−0.0337	−3.82	−0.000477
tactical_front_rel	+0.0330	+3.54	−0.000040
booking_upgrade_reliable	+0.0316	+3.44	+0.000606
booking_upgrade	+0.0279	+2.89	+0.000404
tactical_gain_x_horse_back	+0.0273	+2.87	−0.000444
first_partnership_f	+0.0235	+2.28	−0.000170
appointment_z	+0.0197	+2.16	+0.000520
appointment_deviation	+0.0196	+2.15	+0.000327
tj_market_gap	+0.0191	+2.15	−0.000045
jp_adj_h365	+0.0191	+2.06	+0.000107
change_available	−0.0237	−1.93	−0.000204
booking_state_available	−0.0226	−1.85	−0.000195
jp_h365	+0.0171	+1.79	+0.000229
booking_upgrade_pred_reliable	+0.0148	+1.62	+0.000774

The excluded positive control. Newton unbounded: the column separates winners within race, so the coefficient has no finite maximum-likelihood value. Reported as the best value on a bounded

grid. This column is the race’s own realised relative performance. It is never a candidate; it is here to show the test can detect something when something is there.

Permutation controls.

Control	forward r	permuted
jockey performance state, real	+0.2668	—
jockey performance state, permuted within year	+0.0005	—
jockey performance state, permuted within meeting	+0.0039	—
jockey_changed_f, real against trainer x market-band permutation	−0.0532	−0.0463
returning_f, real against trainer x market-band permutation	+0.0084	−0.0053
booking_upgrade, real against trainer x market-band permutation	+0.0319	+0.0377

Redundancy. Novel share is $1 - R^2$ of a boosted reconstruction fitted before 2022 and scored after. The plan is explicit that novelty is not evidence of usefulness; three studies found the two inversely related.

Feature	novel vs the 28	novel vs jockey_residual
jr_neff	1.0000	1.0000
jp_neff	1.0000	1.0000
tj_share	1.0000	1.0000
tj_available	1.0000	1.0000
stable_first_call	1.0000	1.0000
trial_ride_90d	1.0000	1.0000
jr_trend	0.9940	0.9982
jr_se	0.9685	0.9714
tj_rides_log	0.9648	1.0000
jr_available	0.9454	1.0000
returning_f	0.9224	1.0000
tj_market_gap	0.9006	0.9638
booking_upgrade_pred	0.8864	0.9878
booking_upgrade_pred_reliable	0.8802	0.9812
first_partnership_f	0.8586	0.9986
jockey_changed_f	0.8575	1.0000

10 Market-offset and canonical results

The canonical table is in §1, where it belongs, because it is the campaign’s answer rather than an appendix to it. This section carries the ladder that produced it.

Development scale is used for defects and gross rejection only. That is the plan’s instruction and the programme’s own record supports it emphatically: ten development-to-canonical retentions are now on file across five studies, of which one changed sign and the median is uninformative. No block was ranked against a close competitor on development output.

11 Production feasibility: no finalist to hand over

No block earned a handoff, so none was written. The handoff machinery exists and runs — stages/j09_production_handoff.py generates a candidate manifest, a production card, an expected-columns contract, a research-result record, an implementation prompt and deterministic parity fixtures for any block labelled **production-finalist** or **research-survivor**. Eight blocks

Fig J6. Controls, market offset, and redundancy

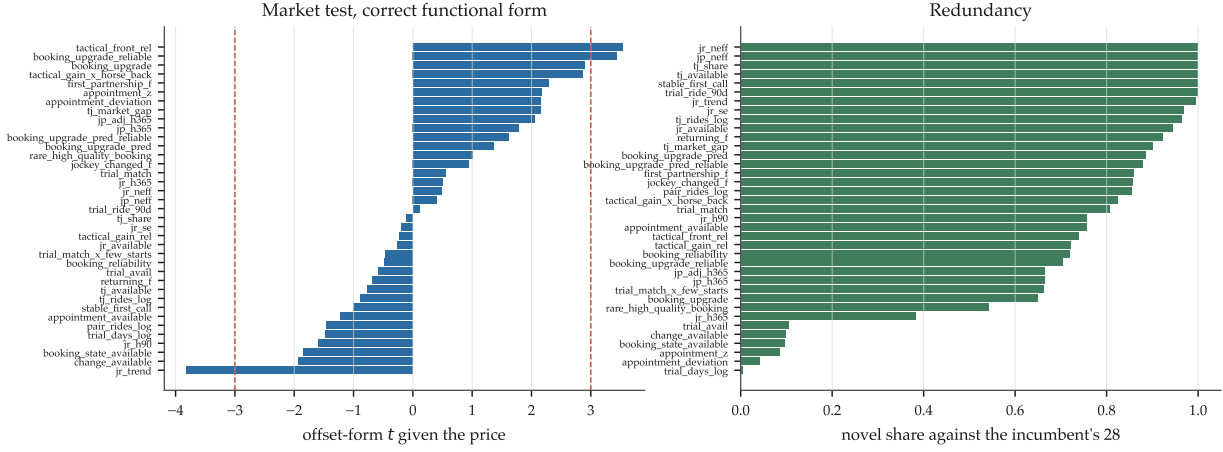


Figure 6: J6. Controls, market offset, and redundancy. Left: the offset-form t statistic for every candidate column, with the old battery's $|t| > 3$ bar marked. Right: novel share against the incumbent's 28 columns.

Fig J7. Canonical confirmation of the advanced blocks

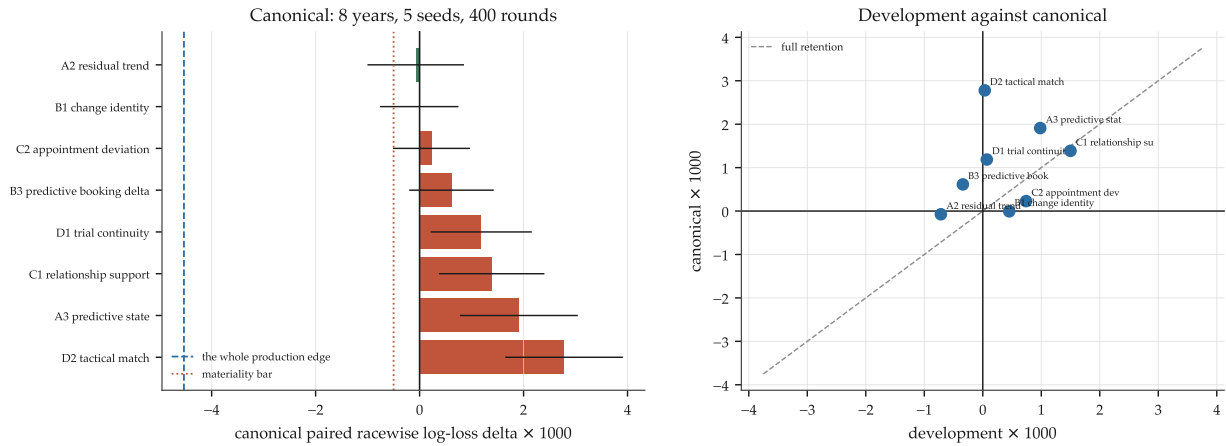


Figure 7: J7. Canonical confirmation. Left: the paired canonical delta per advanced block with its standard error, against the materiality bar and the whole production edge. Right: development against canonical, with the full-retention diagonal.

reached canonical scale and every one was **rejected**, so `artifacts/production-handoff/` contains an index recording that outcome and no candidate directory. Writing a card for a rejected block would misrepresent the evidence; the plan’s instruction not to loosen gates applies to the handoff as much as to the canonical table.

For the record, the feasibility work was done and it is not what stopped anything. Every proposed column is computable from datasets `hkjc-mod` already loads — `aspx-results` for identity and results, `barrier-trial-results` for trial continuity (already loaded for the `bt_*` features), and its own feature table for the market — so no new ingestion would have been required. Nine of the eleven blocks need no cache at all: they are shifts, counts, differences and strictly-prior aggregations that fit beside the existing `jockey_residual` logic in `src/hkjc/features/build.py`. The two exceptions are the predictive-state blocks, which depend on a within-race relative-performance target production does not compute, and the tactical block, whose forward-fitted residualisation is the one case where the plan’s bar for a separate cache module would have been met. Those dependencies are recorded in `stages/j09_production_handoff.py` so that a future campaign reaching a survivor does not have to re-derive them.

12 Limitations, multiple-search accounting, and the prospective requirement

Search accounting. Four waves were predeclared and four were run. Eleven candidate blocks were built against a budget of twelve. Thirty-seven candidate columns were constructed. At most two blocks per wave advanced to canonical scale under a rule fixed before any canonical number was read. No block was split into post-hoc variants, and no gate was moved.

Reused data. Every season through 2026 has been development data in this repository and in the production one. That is the binding limitation on the whole campaign and the reason no label above `production-finalist` is available: retrospective evidence on reused years cannot establish that a feature will help on unseen data, only that it would have helped on seen data under a paired protocol.

What would settle it. A locked implementation, frozen before the fact, scored against time-stamped pre-race prices on races run after the freeze. Until that exists, the correct reading of any survivor here is “worth a production experiment”, not “worth promoting”.

Two specific fragilities. The trial-continuity columns depend on `barrier-trial-results`, whose coverage begins in 2010 and whose jockey field is a display name; a rename would silently reduce the match rate rather than fail. And the tactical block depends on the sectional archive, which covers 2008 onward and not the full production span, so its neutral share is not uniform across the scored years.

13 Final ranked table: every attempted block

Every block attempted, not only those that advanced. Δ_{dev} is the development profile, Δ_{can} the canonical one; a dash means the block did not advance under the predeclared two-per-wave rule.

Wave	Block	cols	Δ_{dev}	Δ_{can}	best $ t $	offset	decision
A	recent market gap	4	+0.001025	—		−1.59	rejected
A	residual trend	2	−0.000716	−0.000075		−3.82	rejected
A	predictive state	3	+0.000982	+0.001911		+2.06	rejected
B	change identity	5	+0.000452	−0.000006		+2.28	rejected
B	booking delta	4	+0.001821	—		+3.44	rejected
B	predictive booking delta	3	−0.000340	+0.000614		−1.85	rejected
C	relationship support	4	+0.001499	+0.001387		−1.00	rejected
C	appointment deviation	4	+0.000740	+0.000229		+2.16	rejected
C	compact combined	3	+0.001640	—		+2.16	rejected
D	trial continuity	5	+0.000069	+0.001186		−1.48	rejected
D	tactical match	3	+0.000034	+0.002779		+3.54	rejected

Advancement, predeclared. At most two blocks per wave, selected on development output alone:

- wave A: advanced A2 residual trend, A3 predictive state of 3 considered.
- wave B: advanced B3 predictive booking delta, B1 change identity of 3 considered.
- wave C: advanced C2 appointment deviation, C1 relationship support of 3 considered.
- wave D: advanced D2 tactical match, D1 trial continuity of 2 considered.

Appendix: provenance

Repositories. Research: `hkjc-research`, this repository. Production: `hkjc-mod`, read-only, at the commit recorded in §2 — which is *not* the plan’s reference commit, and whose working tree was dirty because another session was modifying it.

Stages. `j00_production_baseline`, `j01_data_audit`, `j02_jockey_state`, `j03_booking_change`, `j04_trainer_booking`, `j05_trial_tactics`, `j06_controls_redundancy`, `j07_canonical`, `j08_report_tables`. Each writes tables to `outputs/tables/`, a JSON diagnostic to `outputs/diagnostics/`, figures to `outputs/figures/`, and canonical frames to `outputs/canonical/`.

Seeds. Permutation controls and the tactical gate use 20260902; the paired bootstrap uses 20260902 with 2000 replicates; the canonical harness uses the production seed set (7, 17, 42, 123, 256) at 400 boosting rounds over eight test years.

Distinguishing evidence classes. Source data lives under the production repository and is never written by this one. Generated intermediates live in `cache/` and are rebuildable from the stages. Exploratory evidence is anything in §3–§9. Canonical evidence is only what `outputs/canonical/j07_*` contains. Production confirmation is absent by construction: it can only be produced by a separate `hkjc-mod` session running `hkjc feature-iterate` against its own incumbent, and no such result is imported here.